

Smart Rule-Based Approval Engine

1. Problem Context

Organizations process multiple approval requests daily:

- Expense Claims
- Leave Requests

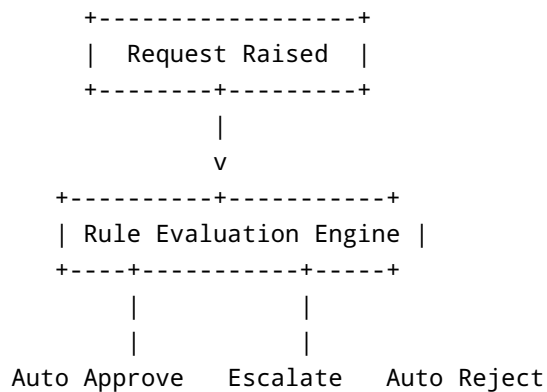
Manual approvals create:

- Delays
- Manager overload
- Policy inconsistency
- Lack of transparency

Solution: Automate approvals using rule based decisioning.

2. System Objective

Evaluate each request automatically and decide:



3. Actors & Hierarchy

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graph TD; Admin --> Manager;
```

The hierarchy diagram shows "Admin" at the top, with a vertical line connecting it to "Manager" below it.

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Employee

Responsibilities

- Employee → Create requests
- Manager → Review escalations
- Admin → Final authority + Rule control

4. Employee Grades

System supports 3 grades:

Grade	Expense Limit	Leave Limit
G1	₹10,000	30 Days
G2	₹25,000	36 Days
G3	₹50,000	45 Days

Notes

- Limits are annual
- Can be used anytime
- Bulk usage allowed

5. Request Categories

1. Expense
2. Leave

Configurable via Admin rules.

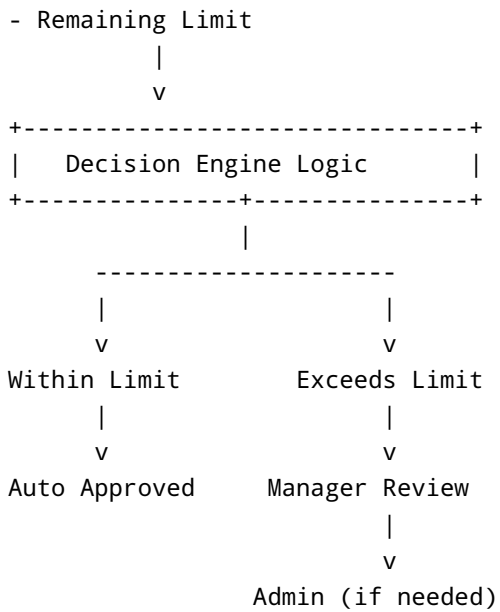
6. End-to-End Request Flow

Employee Submits Request

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System Fetches:

- Grade
- Category
- Annual Usage



7. Approval Outcomes

Scenario	Action
Within threshold	Auto Approval
Slight exceed	Manager Review
Major violation	Admin Decision

8. Request Lifecycle

Submitted → Under Review → Approved
 → Rejected
 → Auto Approved

Each state is tracked.

9. Transparency Layer

Every request shows:

- Applied rule
- Approval reason
- Escalation cause

- Approver comments

Ensures audit readiness.

10. Admin Control Panel

Rule Management

- Create rules
- Modify limits
- Delete rules

Oversight

- View all requests
- Override decisions
- Approve escalations

Analytics

- Rule usage
 - Approval patterns
 - Request statistics
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11. Acceptance Coverage

- ✓ Grade based approvals
 - ✓ Category rules
 - ✓ Auto approval within limits
 - ✓ Manager escalation
 - ✓ Admin final authority
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